

Lecture 4: Vector-Valued Functions

Background & Motivation

We now shift from static geometry to motion. A vector-valued function $\mathbf{r}(t)$ traces a curve through space as the parameter t varies—think of it as the position of a particle at time t . This chapter connects the vector algebra from Chapter 1 to calculus: we'll differentiate and integrate these functions component by component, leading to tangent vectors, arc length, curvature, and projectile motion.

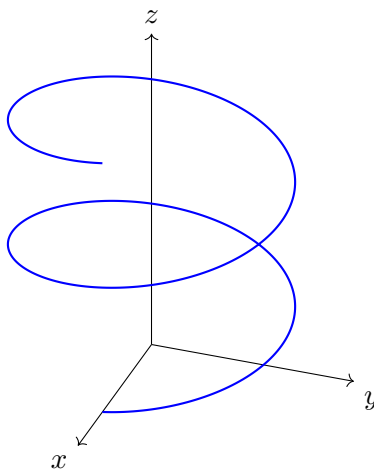
1. Parametric Curves in $\mathbb{R}^3$

Definition 1: Vector-Valued Function (VVF)

A VVF maps a scalar t to a vector:

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}} + h(t)\hat{\mathbf{k}}$$

- f, g, h are the **component functions**
- As t varies, the tip of $\mathbf{r}(t)$ traces a **space curve**



Helix $\mathbf{r}(t) = \langle R \cos t, R \sin t, ct \rangle$ spiraling around the z -axis. As t increases, $\mathbf{r}(t)$ rotates in the xy -plane while rising linearly in z .

Common Parametric Curves

- **Line:** $\mathbf{r}(t) = \mathbf{r}_0 + t \mathbf{d}$
- **Circle** (radius R , xy -plane): $\mathbf{r}(t) = \langle R \cos t, R \sin t, 0 \rangle$, $t \in [0, 2\pi]$
- **Helix** (radius R , rising rate c): $\mathbf{r}(t) = \langle R \cos t, R \sin t, ct \rangle$
- **Ellipse:** $\mathbf{r}(t) = \langle a \cos t, b \sin t, 0 \rangle$

Example 1: Identifying a Helix

Describe the curve $\mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, 2t \rangle$ for $t \geq 0$.

1. Check: $x^2 + y^2 = 9 \cos^2 t + 9 \sin^2 t = 9$ – lies on the cylinder $x^2 + y^2 = 9$.
2. The z -component $z = 2t$ increases linearly.
3. Result: a **helix** of radius 3, rising at rate 2 per radian, winding around the z -axis.

2. Derivatives of VVFs

Definition 2: Derivative of a VVF

The derivative $\mathbf{r}'(t)$ is the instantaneous rate of change of the position vector, obtained by differentiating each component separately. Geometrically it is the **velocity vector**: it points in the direction the curve is moving at that instant, and its magnitude is the speed at which the particle is traversing the curve.

$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$$

- $\mathbf{r}'(t)$ is a **tangent vector** to the curve at $\mathbf{r}(t)$
- $\|\mathbf{r}'(t)\|$ is the **speed**

Derivative Rules for VVFs

- Scalar product rule: $\frac{d}{dt}[f(t)\mathbf{r}(t)] = f'(t)\mathbf{r}(t) + f(t)\mathbf{r}'(t)$
- Dot product rule: $\frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{s}(t)] = \mathbf{r}'(t) \cdot \mathbf{s}(t) + \mathbf{r}(t) \cdot \mathbf{s}'(t)$
- Cross product rule: $\frac{d}{dt}[\mathbf{r}(t) \times \mathbf{s}(t)] = \mathbf{r}'(t) \times \mathbf{s}(t) + \mathbf{r}(t) \times \mathbf{s}'(t)$
- Chain rule: $\frac{d}{dt}[\mathbf{r}(g(t))] = \mathbf{r}'(g(t)) \cdot g'(t)$

Example 2: Derivative and Analysis at a Point

Let $\mathbf{r}(t) = \langle t^3, \sin t, e^t \rangle$. Find $\mathbf{r}'(t)$ and analyze at $t = \pi$.

1. Derivative: $\mathbf{r}'(t) = \langle 3t^2, \cos t, e^t \rangle$
2. At $t = \pi$:
 - Position: $\mathbf{r}(\pi) = \langle \pi^3, 0, e^\pi \rangle$
 - Tangent vector: $\mathbf{r}'(\pi) = \langle 3\pi^2, -1, e^\pi \rangle$
3. Component analysis: x -comp > 0 (increasing), y -comp < 0 (decreasing), z -comp > 0 (increasing)
4. Speed: $\|\mathbf{r}'(\pi)\| = \sqrt{9\pi^4 + 1 + e^{2\pi}}$

3. Integrals of VVFs

Definition 3: Integral of a VVF

Also **component-wise**:

$$\int \mathbf{r}(t) dt = \left\langle \int f(t) dt, \int g(t) dt, \int h(t) dt \right\rangle + \mathbf{C}$$

– $\mathbf{C} = \langle C_1, C_2, C_3 \rangle$ is a **vector constant** of integration

$$\int_a^b \mathbf{r}(t) dt = \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle$$

Example 3: Definite Integral of a VVF

Evaluate $\int_0^1 \langle 3t^2, 2 \sin t, e^{2t} \rangle dt$.

1. x -component: $\int_0^1 3t^2 dt = [t^3]_0^1 = 1$
2. y -component: $\int_0^1 2 \sin t dt = [-2 \cos t]_0^1 = 2 - 2 \cos 1$
3. z -component: $\int_0^1 e^{2t} dt = \left[\frac{e^{2t}}{2} \right]_0^1 = \frac{e^2 - 1}{2}$
4. Result: $\left\langle 1, 2 - 2 \cos 1, \frac{e^2 - 1}{2} \right\rangle$

4. Tangent Lines and Curves on Surfaces

Tangent Line at $t = t_0$

$$\mathbf{L}(s) = \mathbf{r}(t_0) + s \mathbf{r}'(t_0)$$

– Point of tangency: $\mathbf{r}(t_0)$ – Direction: $\mathbf{r}'(t_0)$

– To show a curve lies on a surface: substitute the components into the surface equation and verify

Example 4: Tangent Line + Curve on a Surface

The curve $\mathbf{r}(t) = \langle e^t, e^{-t}, t \rangle$. Show it lies on $xy = 1$ and find the tangent line at $t = 0$.

1. On the surface: $x \cdot y = e^t \cdot e^{-t} = e^0 = 1$ ✓ for all t .
2. At $t = 0$: $\mathbf{r}(0) = \langle 1, 1, 0 \rangle$
3. Tangent vector: $\mathbf{r}'(t) = \langle e^t, -e^{-t}, 1 \rangle \implies \mathbf{r}'(0) = \langle 1, -1, 1 \rangle$
4. Tangent line: $\mathbf{L}(s) = \langle 1 + s, 1 - s, s \rangle$

Show all work for full credit.

Practice Problems

Problem 1. If $\mathbf{r}(t) = \langle t^2, \ln(t+1), \cos t \rangle$, find $\mathbf{r}'(t)$ and $\mathbf{r}'(0)$.

Answer: $\mathbf{r}'(t) = \langle 2t, \frac{1}{t+1}, -\sin t \rangle$, $\mathbf{r}'(0) = \langle 0, 1, 0 \rangle$

Problem 2. Evaluate $\int_0^\pi \langle \cos t, \sin t, 1 \rangle dt$.

Answer: $\langle 0, 2, \pi \rangle$

Problem 3. A particle has velocity $\mathbf{v}(t) = \langle 2t, 3, -1 \rangle$ and $\mathbf{r}(0) = \langle 1, 0, 5 \rangle$. Find $\mathbf{r}(t)$.

Answer: $\mathbf{r}(t) = \langle t^2 + 1, 3t, 5 - t \rangle$

Problem 4. Show that $\mathbf{r}(t) = \langle 2 \cos t, 3 \sin t, t \rangle$ lies on the elliptic cylinder $\frac{x^2}{4} + \frac{y^2}{9} = 1$. Find the tangent line at $t = \pi/2$.

Answer: tangent line = $\langle -2s, 3, \pi/2 + s \rangle$

Problem 5. Prove: if $\|\mathbf{r}(t)\| = c$ (constant), then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$ for all t .

Hint: differentiate $\mathbf{r}(t) \cdot \mathbf{r}(t) = c^2$.

$$\frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{r}(t)] = \frac{d}{dt}[c^2] = 0$$

By dot product rule: $\mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) = 0 \implies 2\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$. $\square$

Interpretation: a curve on a sphere has velocity always perpendicular to the position vector.